

Bitwise differential trail through **LBlock** found by SAT

CiVerLy

July 24, 2026

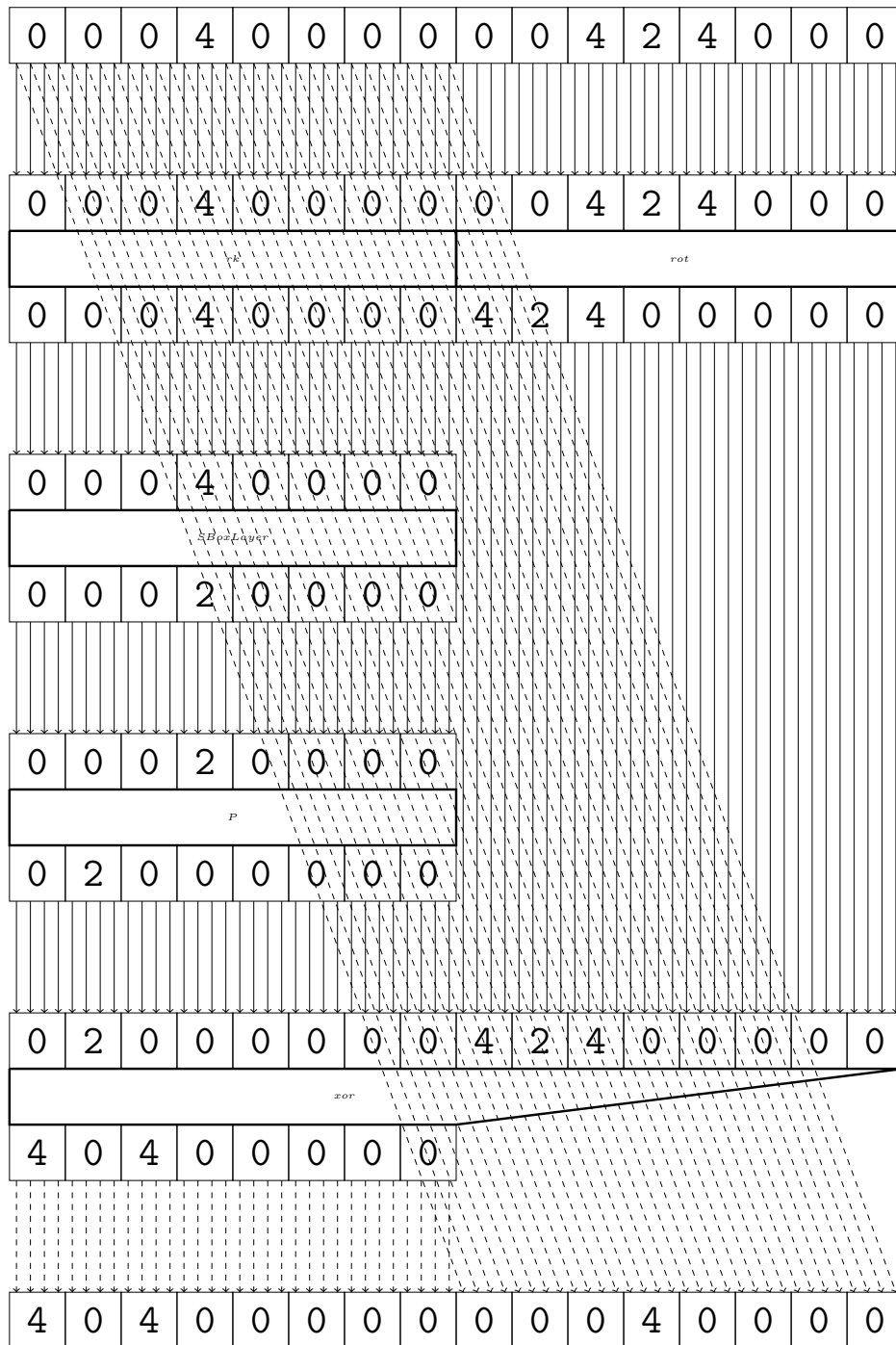
1 LBlock

Maximal differential probability: 2^{-62}

0	0	0	4	0	0	0	0	0	0	4	2	4	0	0	0	
0	0	0	4	0	0	0	0	0	0	4	2	4	0	0	0	
4	0	4	0	0	0	0	0	0	0	0	4	0	0	0	0	
4	0	4	0	0	0	0	0	0	0	0	4	0	0	0	0	
0	4	4	2	0	0	0	0	4	0	4	0	0	0	0	0	
0	4	4	2	0	0	0	0	4	0	4	0	0	0	0	0	
0	5	0	6	0	0	4	0	0	4	4	2	0	0	0	0	
0	5	0	6	0	0	4	0	0	4	4	2	0	0	0	0	
0	0	0	0	0	0	0	0	0	5	0	6	0	0	4	0	
0	0	0	0	0	0	0	0	0	5	0	6	0	0	4	0	
0	6	0	0	4	0	0	5	0	0	0	0	0	0	0	0	
0	6	0	0	4	0	0	5	0	0	0	0	0	0	0	0	
1	0	0	0	0	a	c	0	0	6	0	0	4	0	0	5	
1	0	0	0	0	a	c	0	0	6	0	0	4	0	0	5	
0	0	b	0	2	5	0	0	1	0	0	0	a	c	0	0	
0	0	b	0	2	5	0	0	1	0	0	0	a	c	0	0	
0	0	0	0	0	0	0	0	0	0	b	0	2	5	0	0	
0	0	0	0	0	0	0	0	0	0	b	0	2	5	0	0	
b	0	2	5	0	0	0	0	0	0	0	0	0	0	0	0	
b	0	2	5	0	0	0	0	0	0	0	0	0	0	0	0	
0	2	2	1	0	0	0	0	b	0	2	5	0	0	0	0	
0	2	2	1	0	0	0	0	b	0	2	5	0	0	0	0	
0	0	0	1	0	0	0	0	2	2	1	0	0	0	0	0	
0	0	0	1	0	0	0	0	2	2	1	0	0	0	0	0	
2	0	0	0	0	0	0	0	0	0	0	0	1	0	0	b	0
2	0	0	0	0	0	0	0	0	0	0	0	1	0	0	b	0
0	1	2	0	b	0	0	0	2	0	0	0	0	0	0	0	0
0	1	2	0	b	0	0	0	2	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	0	0	1	2	0	b	0	0	0	0
2	0	0	1	0	0	0	0	0	1	2	0	b	0	0	0	0
0	1	2	0	b	0	0	0	2	0	0	1	0	0	0	0	0
0	1	2	0	b	0	0	0	2	0	0	1	0	0	0	0	0
0	1	2	0	b	0	0	0	2	0	0	1	0	0	0	0	0
0	1	2	0	b	0	0	0	2	0	0	1	0	0	0	0	0

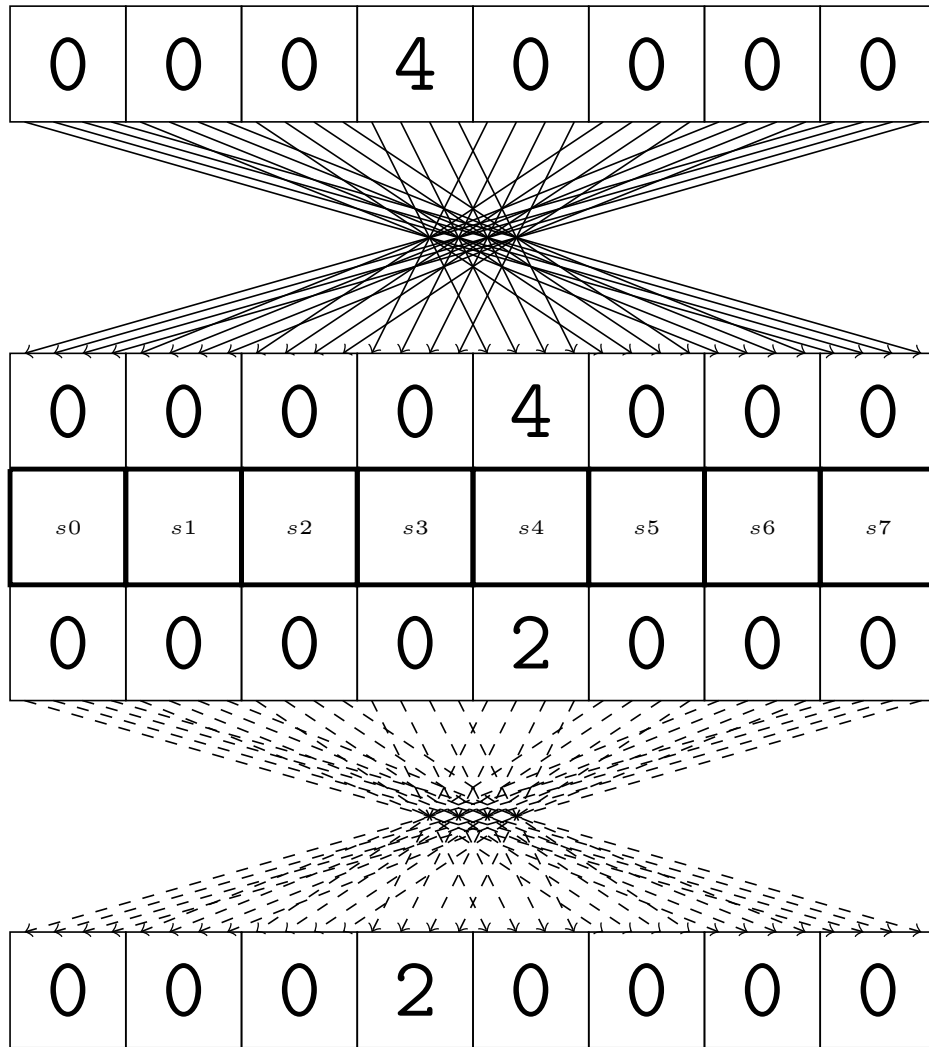
2 LBlock_round

Maximal differential probability: $2^{-2.0}$



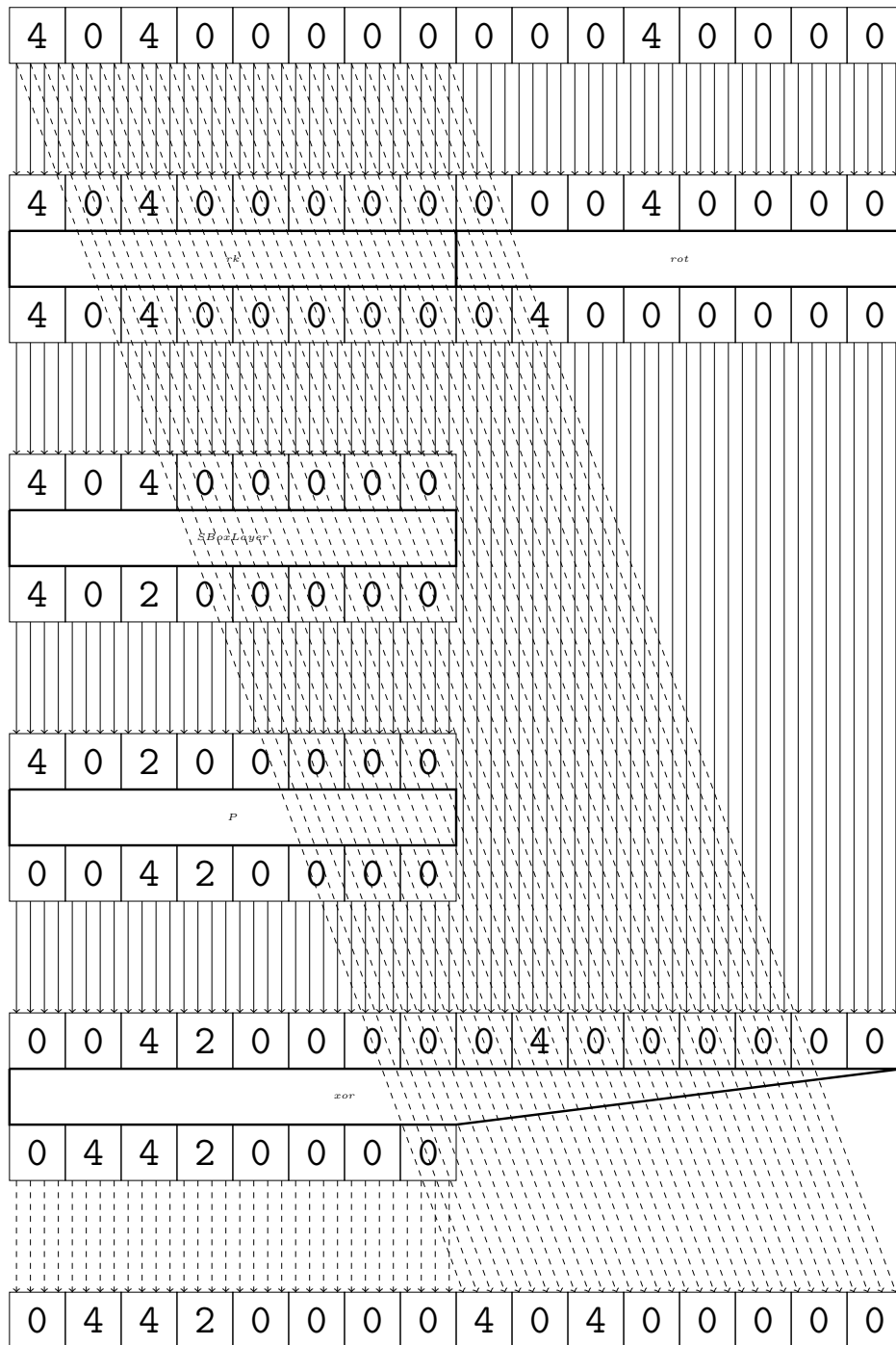
3 SBoxLayer

Maximal differential probability: $2^{-2.0}$



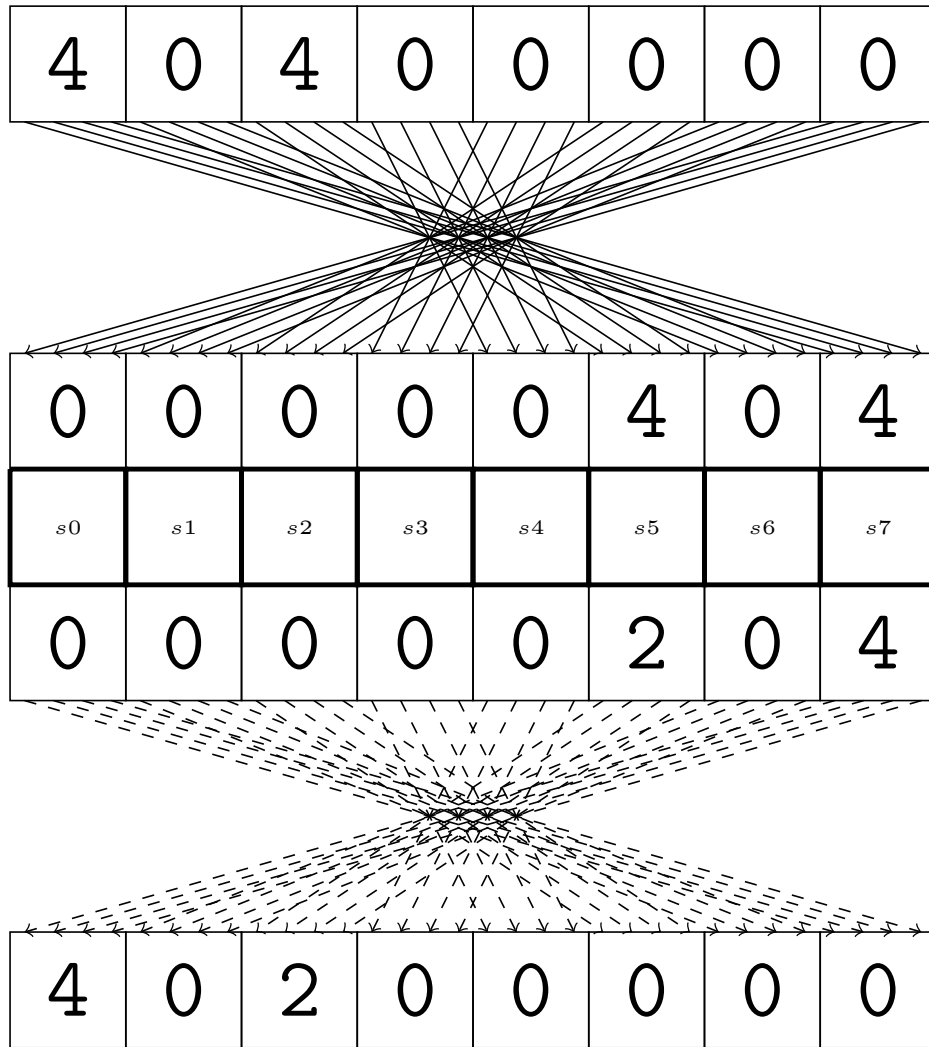
4 LBlock_round

Maximal differential probability: $2^{-4.0}$



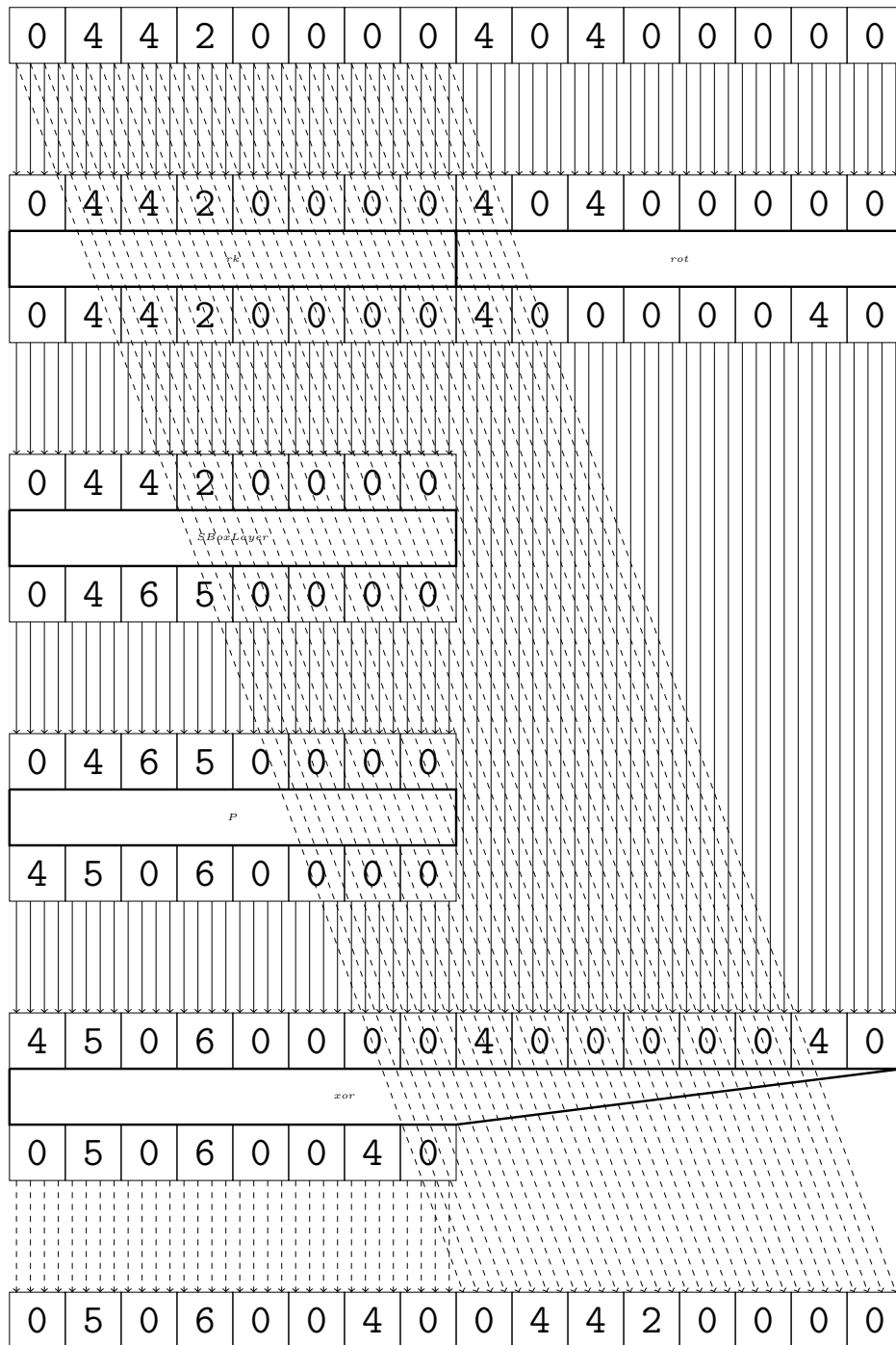
5 SBoxLayer

Maximal differential probability: $2^{-4.0}$



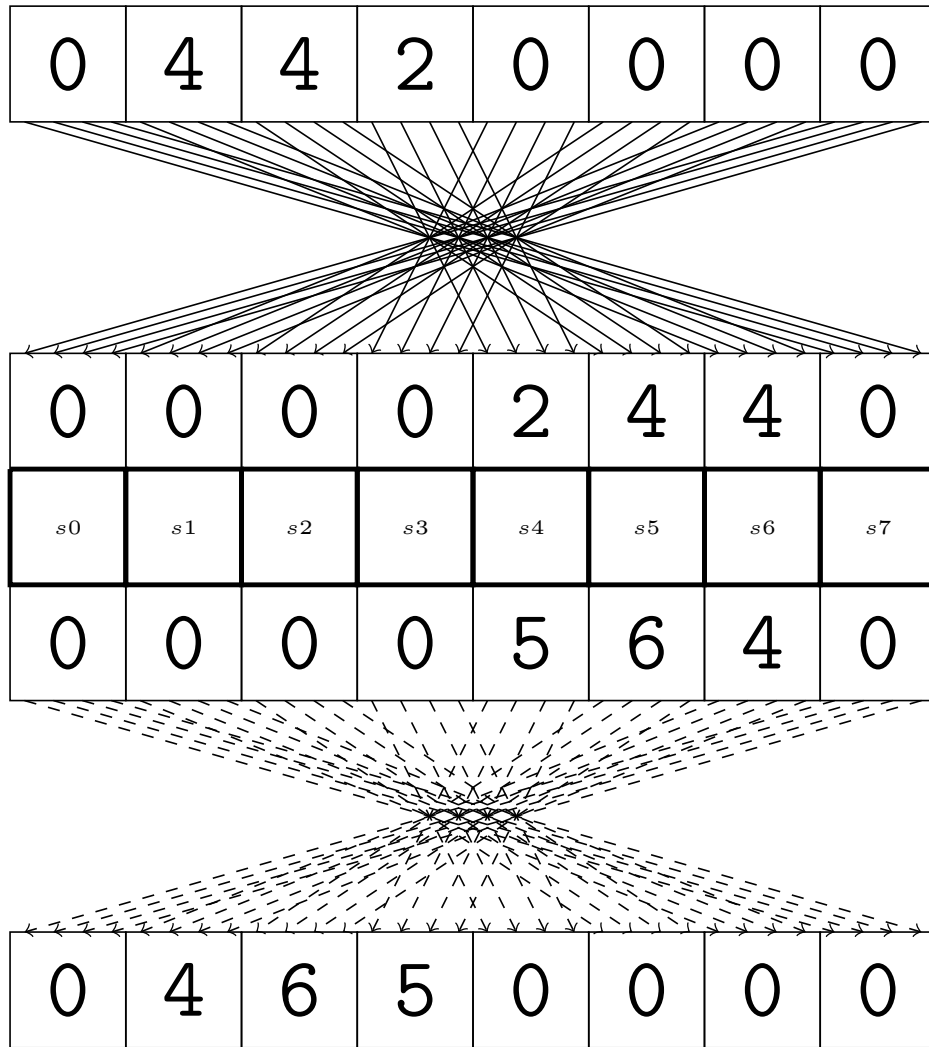
6 LBlock_round

Maximal differential probability: $2^{-6.0}$



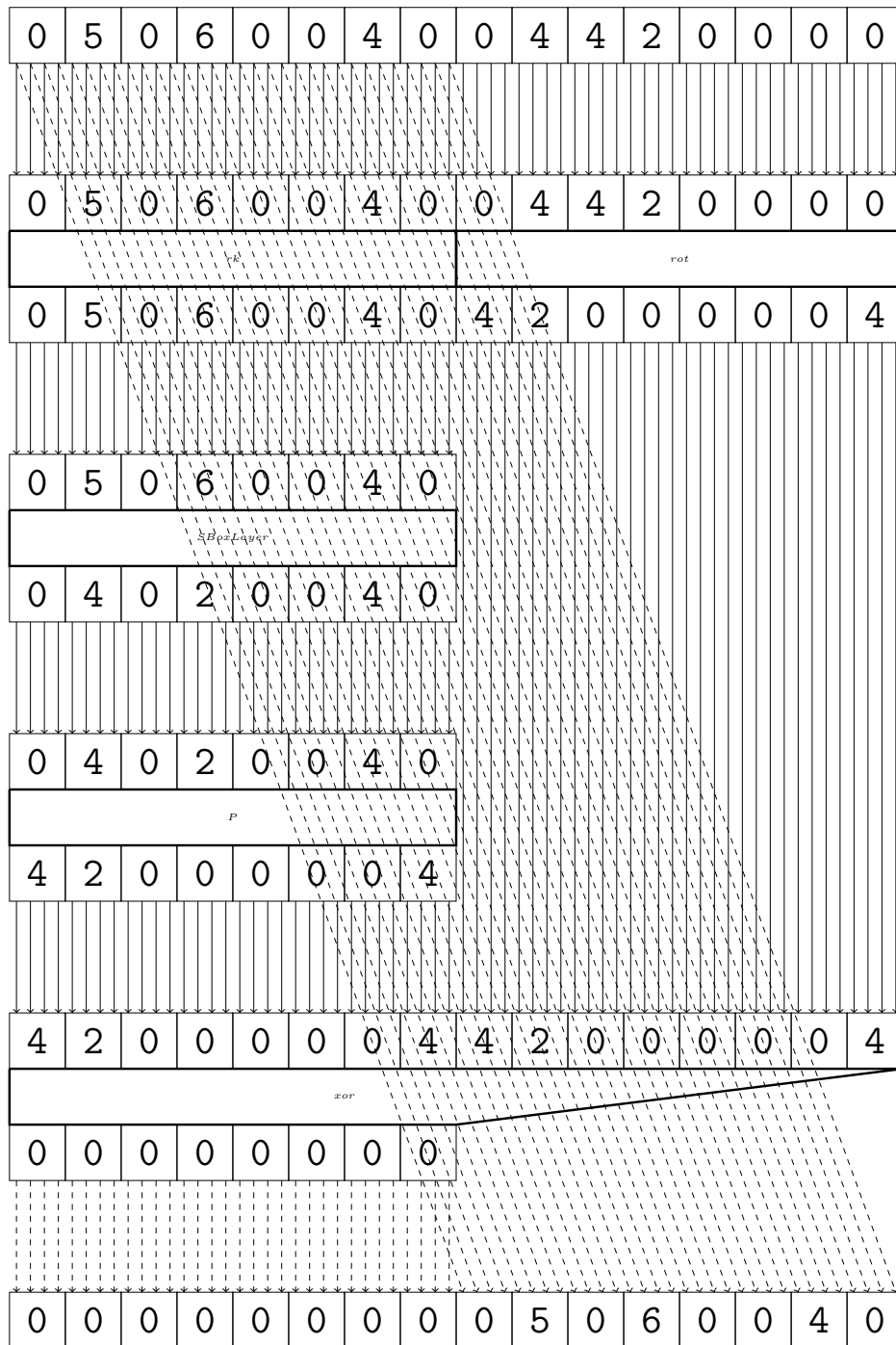
7 SBoxLayer

Maximal differential probability: $2^{-6.0}$



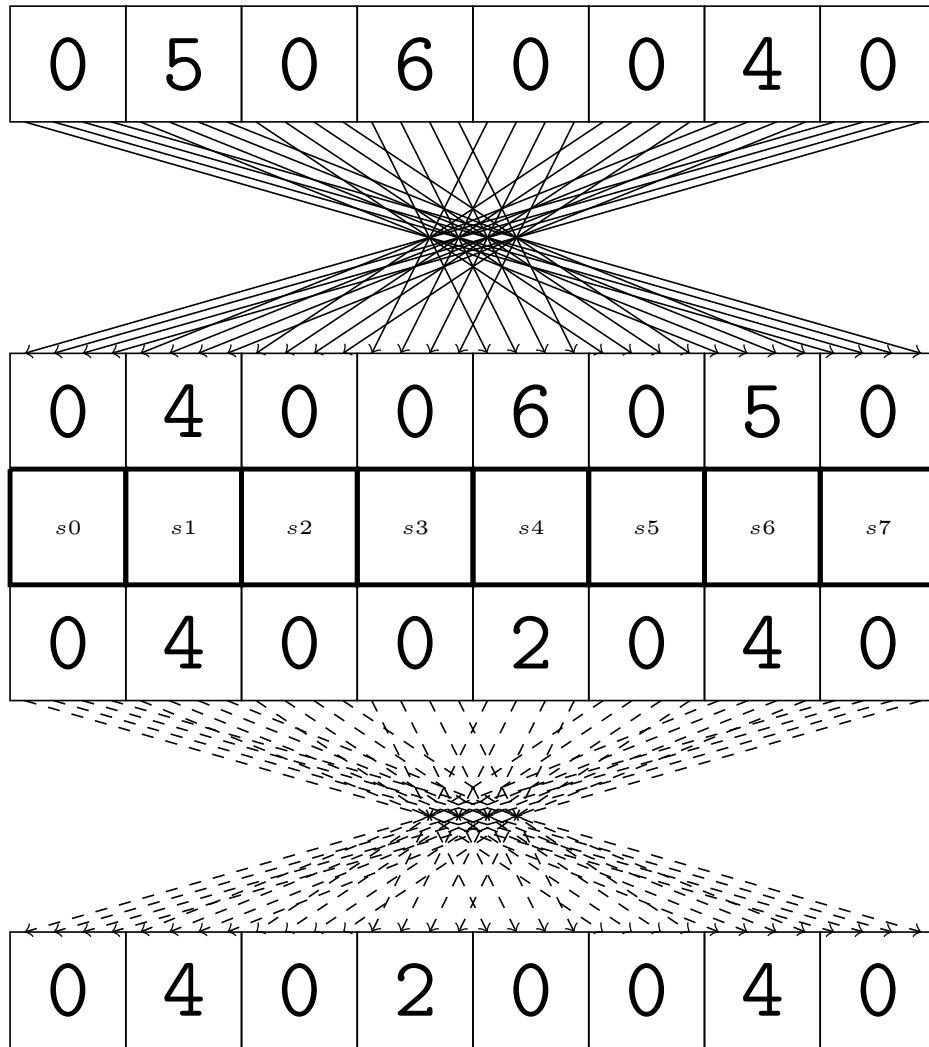
8 LBlock_round

Maximal differential probability: $2^{-6.0}$



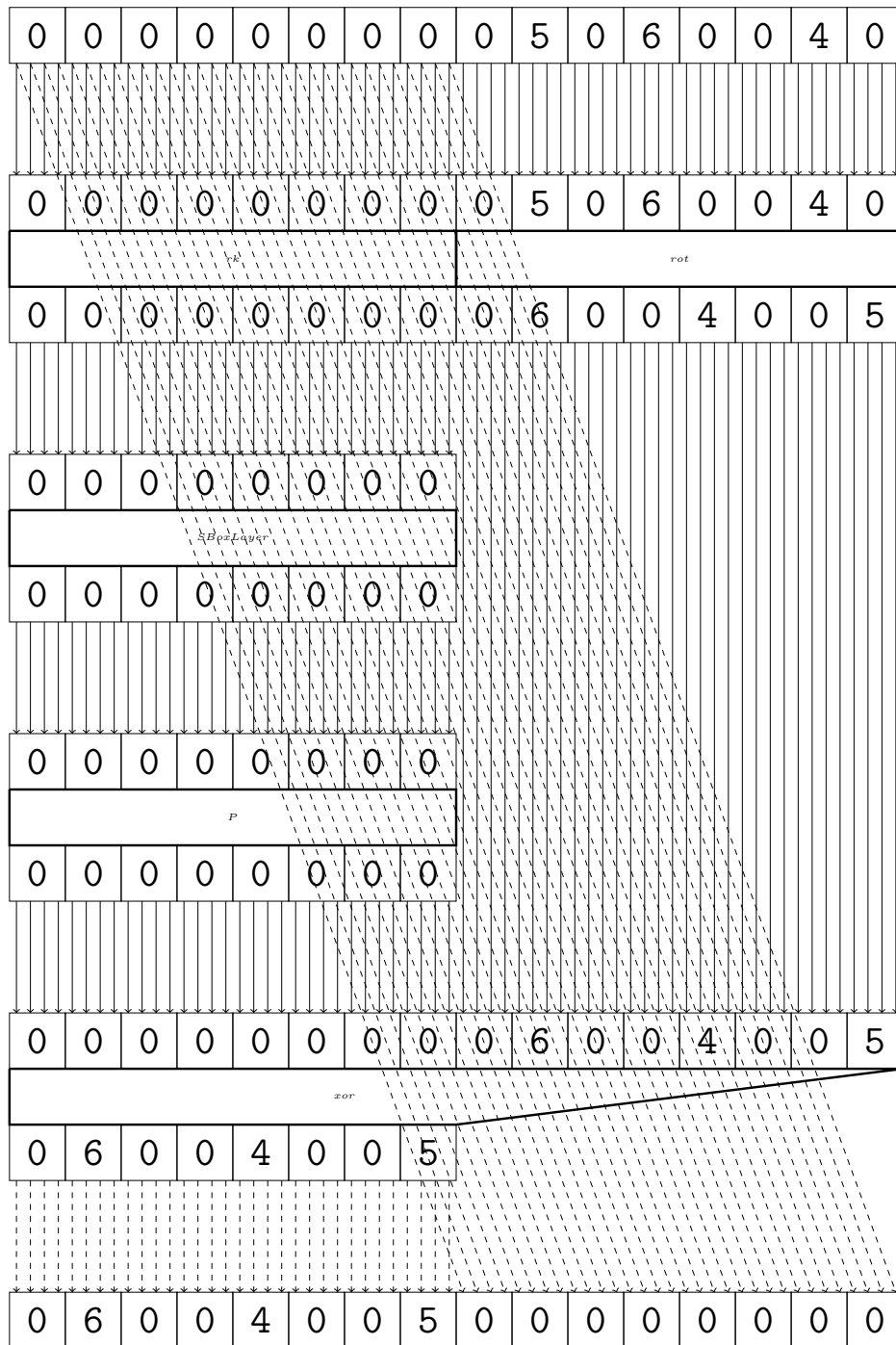
9 SBoxLayer

Maximal differential probability: $2^{-6.0}$



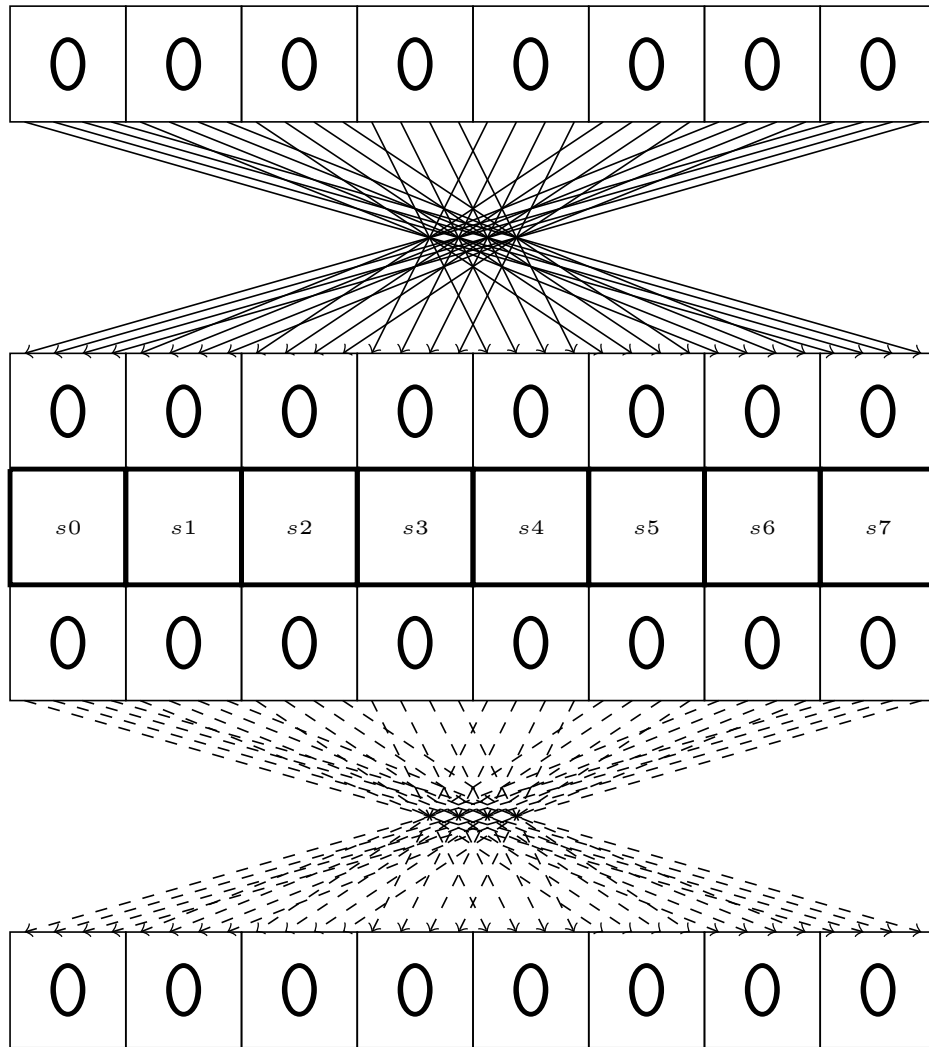
10 LBlock_round

Maximal differential probability: $2^{-0.0}$



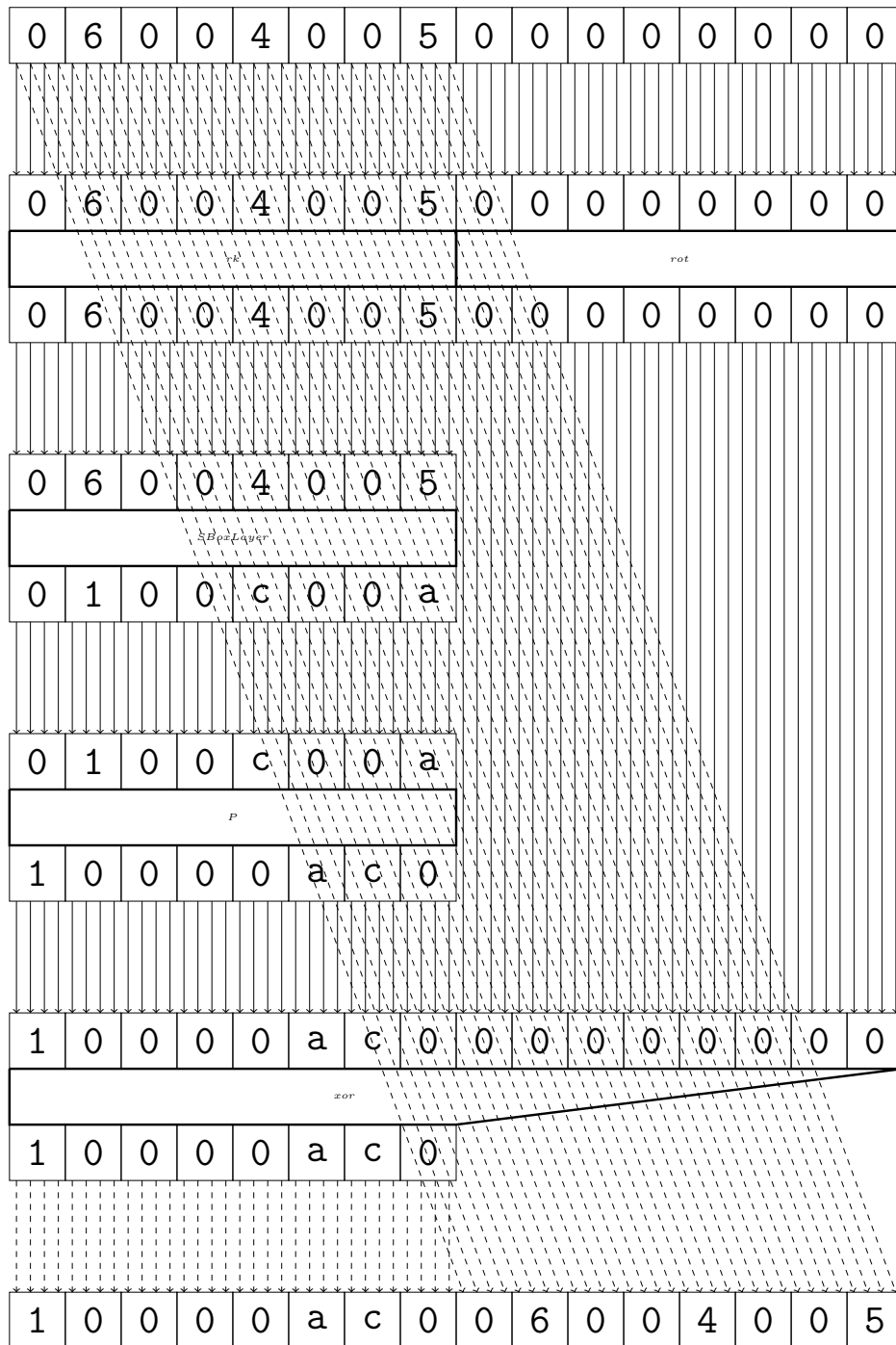
11 SBoxLayer

Maximal differential probability: $2^{-0.0}$



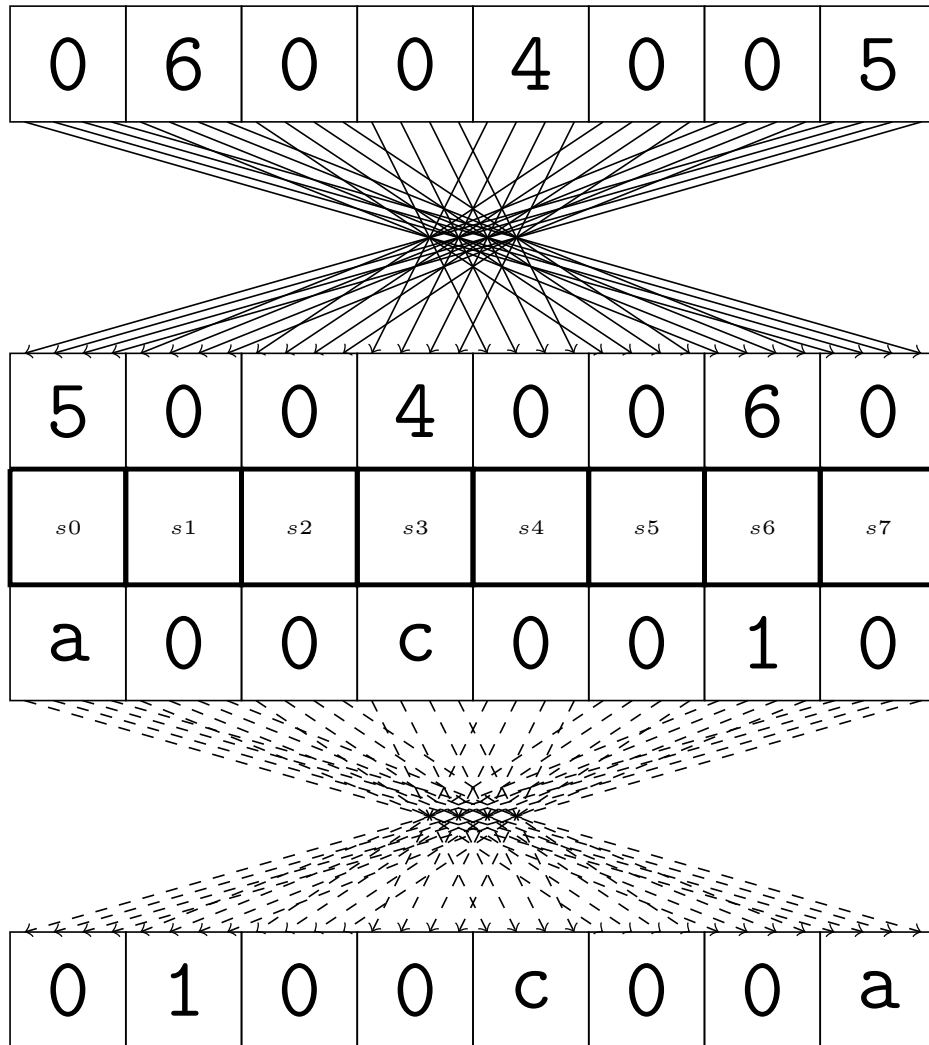
12 LBlock_round

Maximal differential probability: $2^{-6.0}$



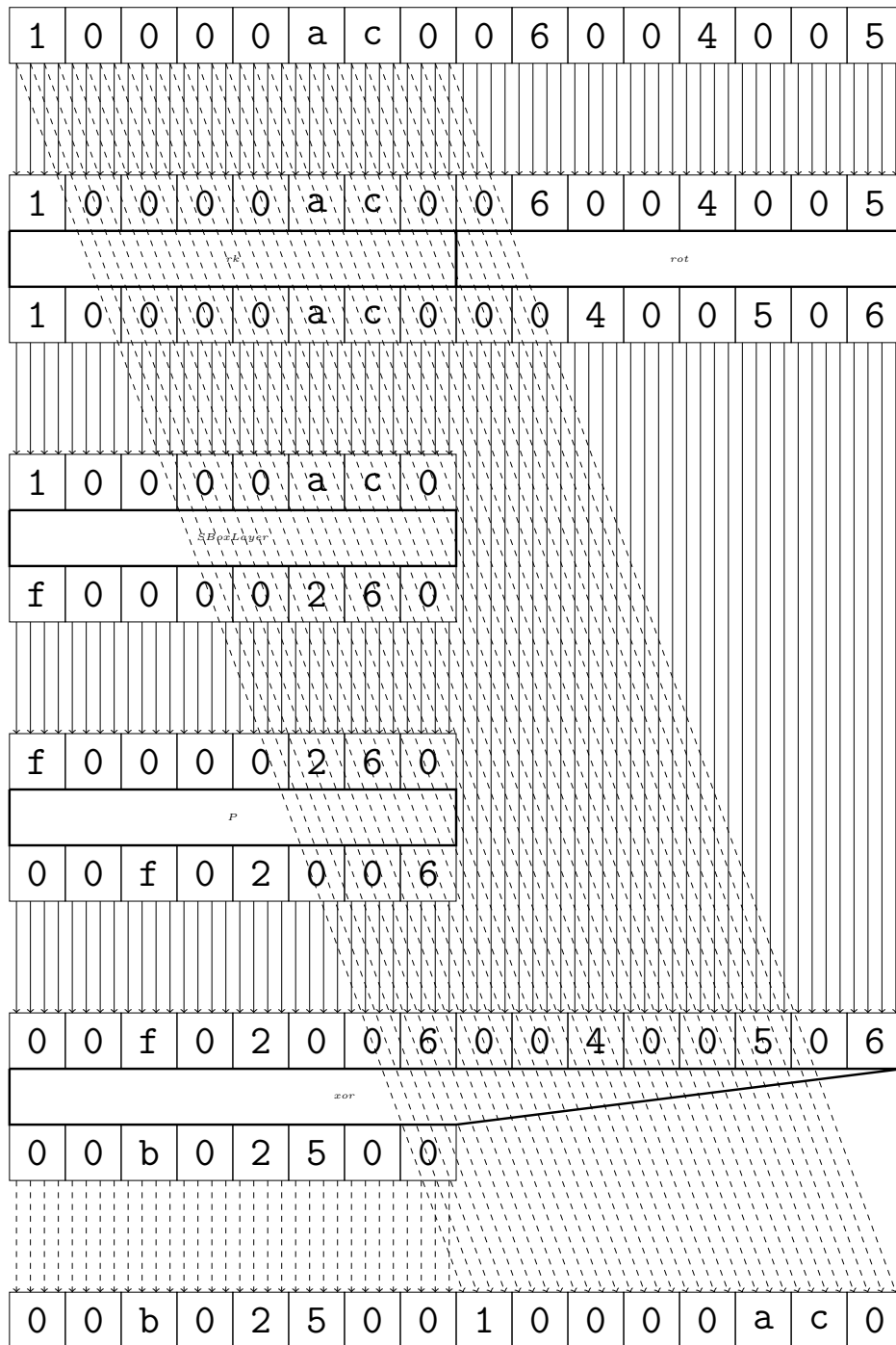
13 SBoxLayer

Maximal differential probability: $2^{-6.0}$



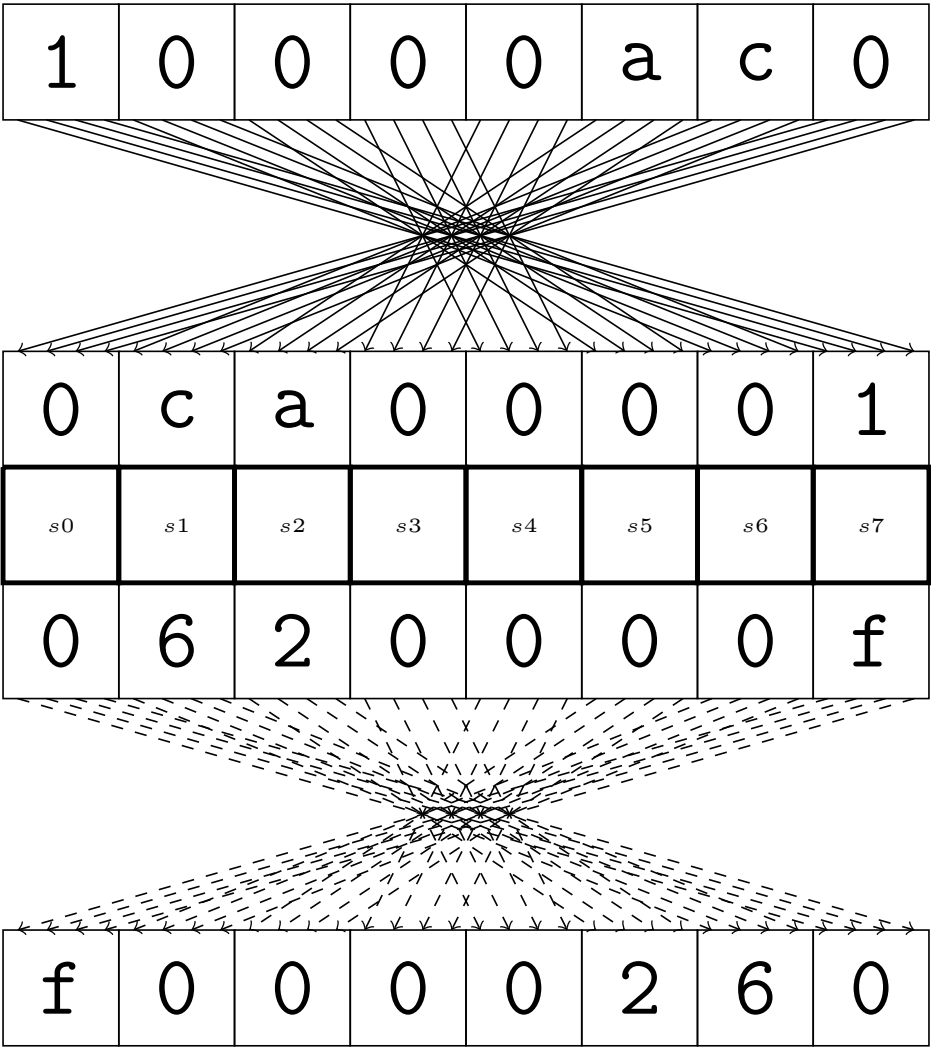
14 LBlock_round

Maximal differential probability: $2^{-8.0}$



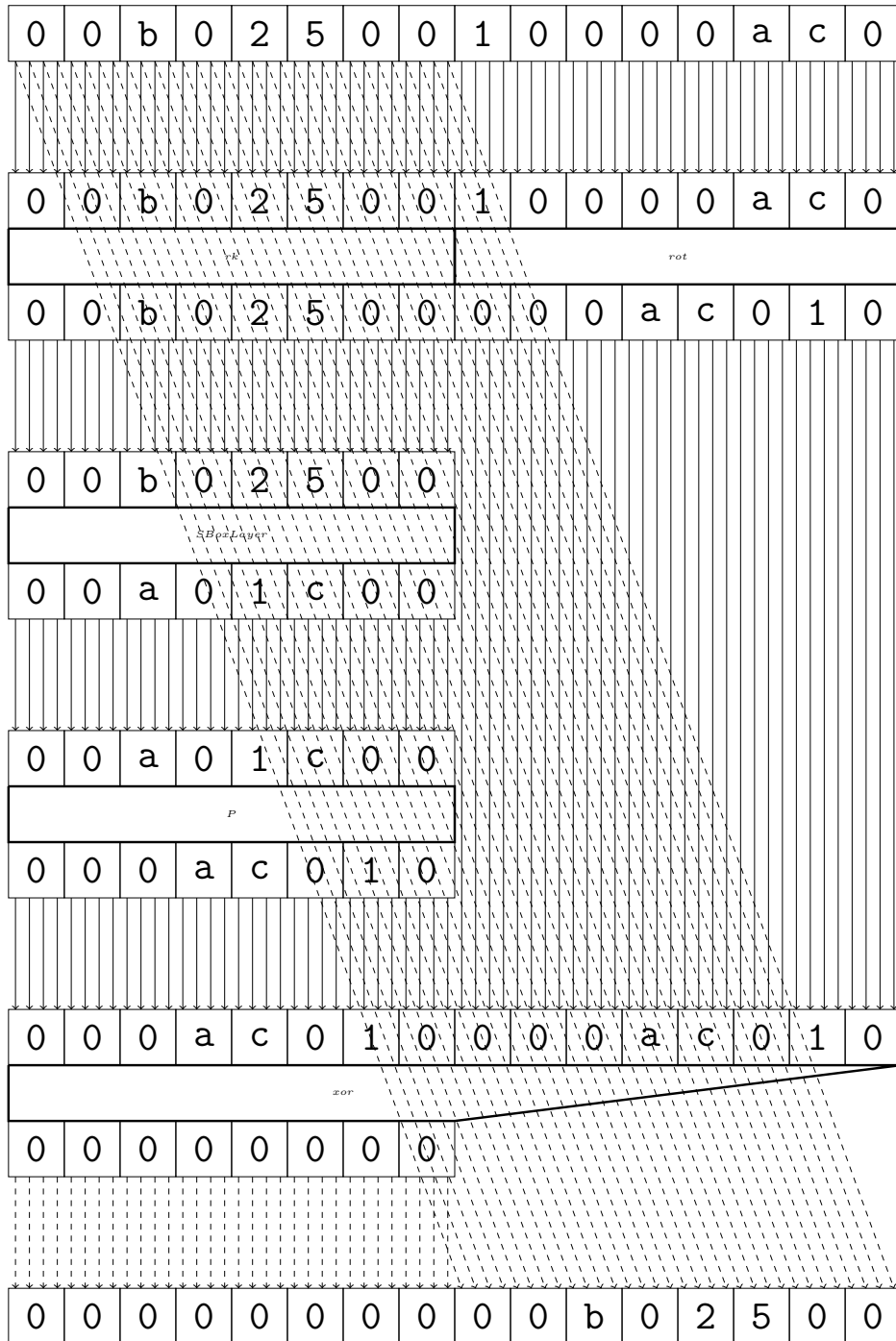
15 SBoxLayer

Maximal differential probability: $2^{-8.0}$



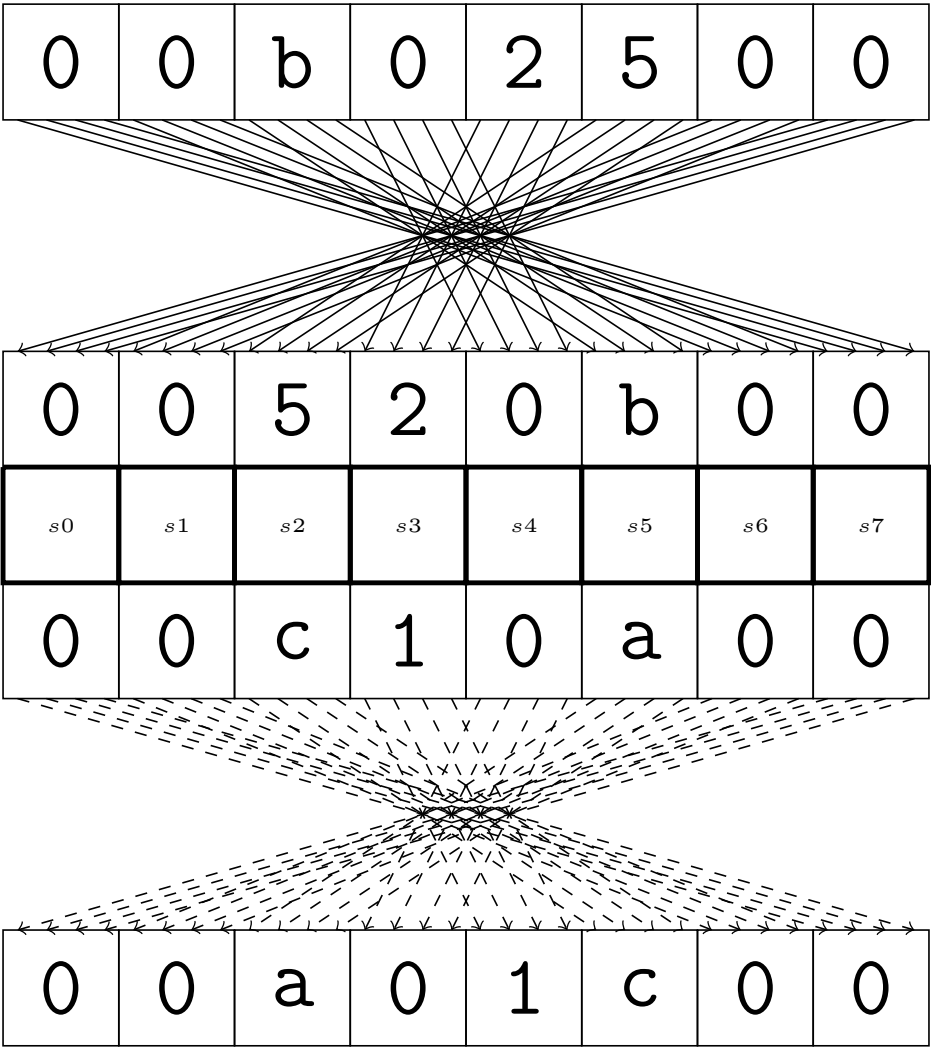
16 LBlock_round

Maximal differential probability: $2^{-6.0}$

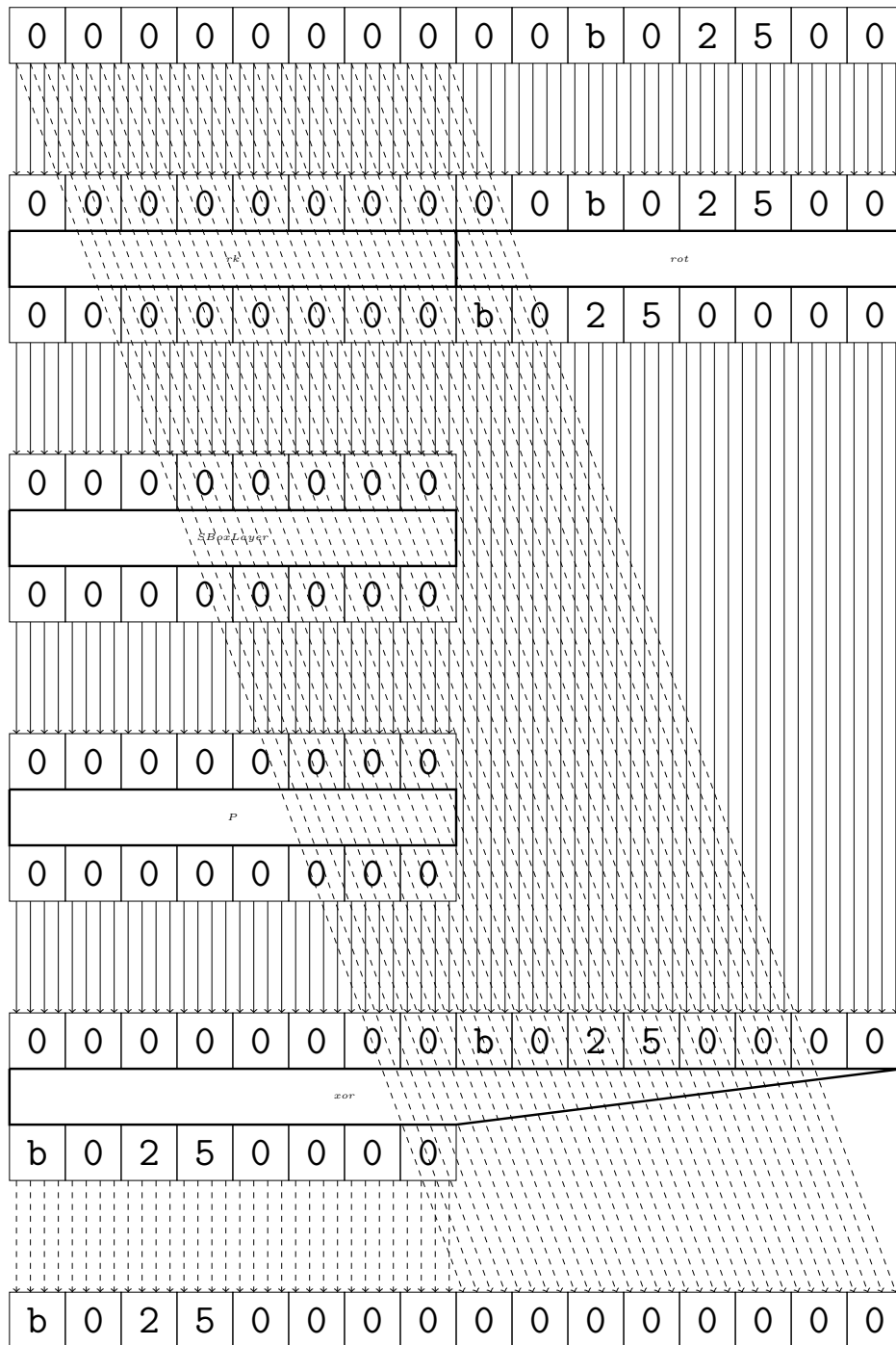


17 SBoxLayer

Maximal differential probability: $2^{-6.0}$

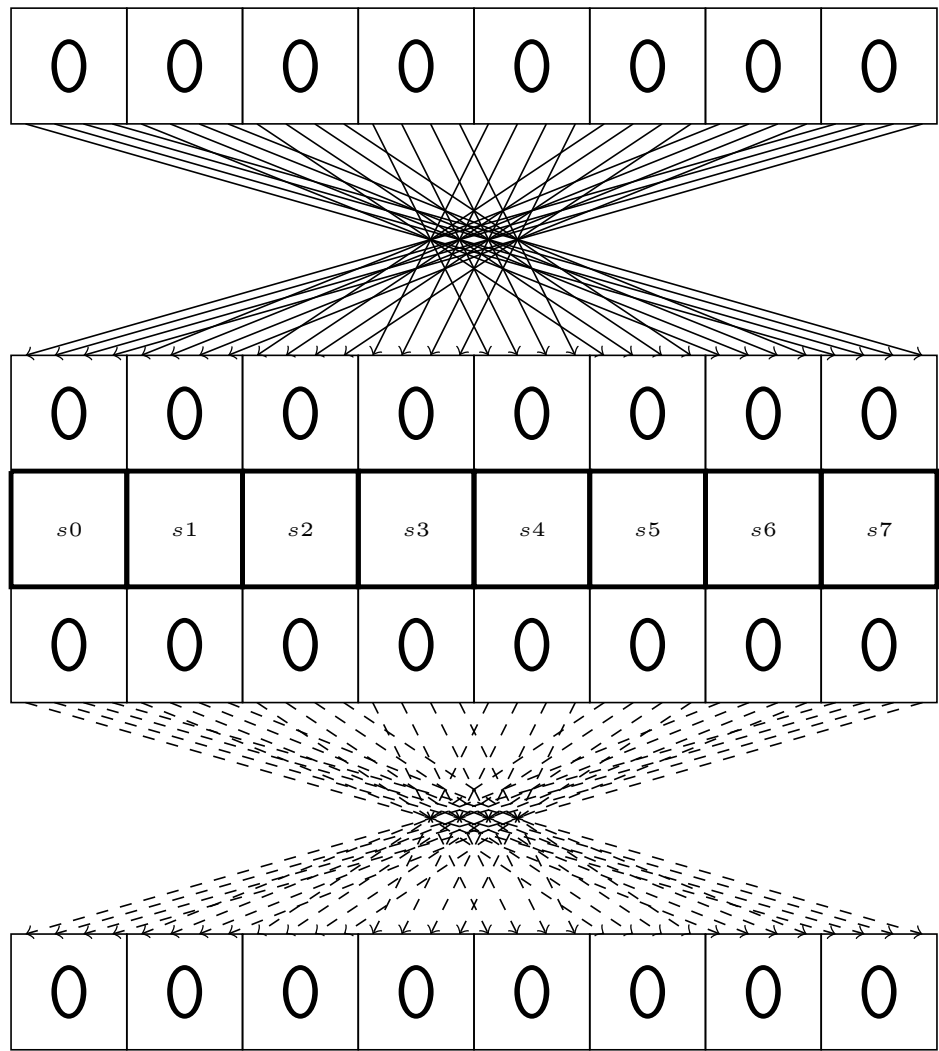


18 LBlock_round

Maximal differential probability: $2^{-0.0}$ 

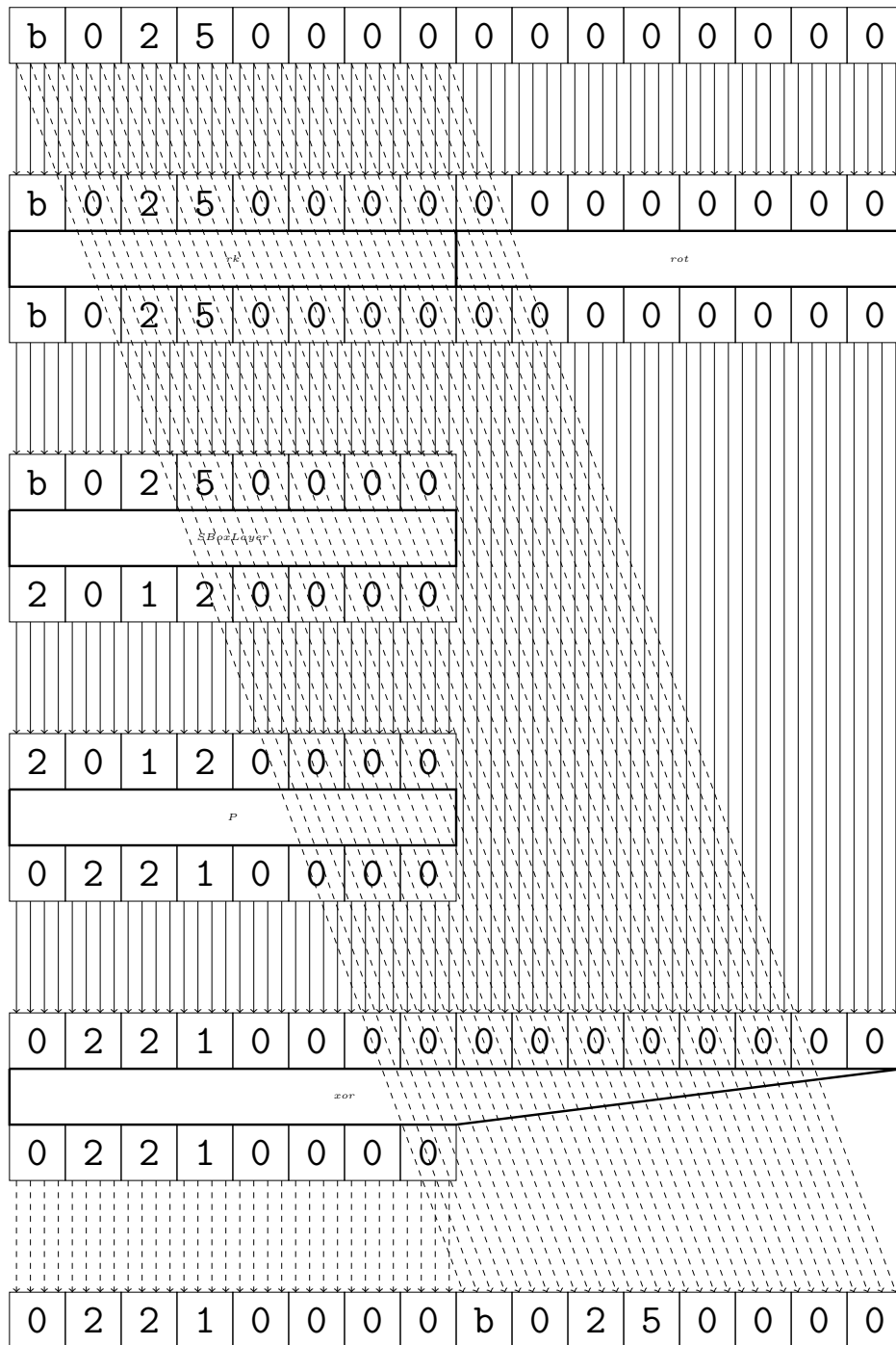
19 SBoxLayer

Maximal differential probability: $2^{-0.0}$



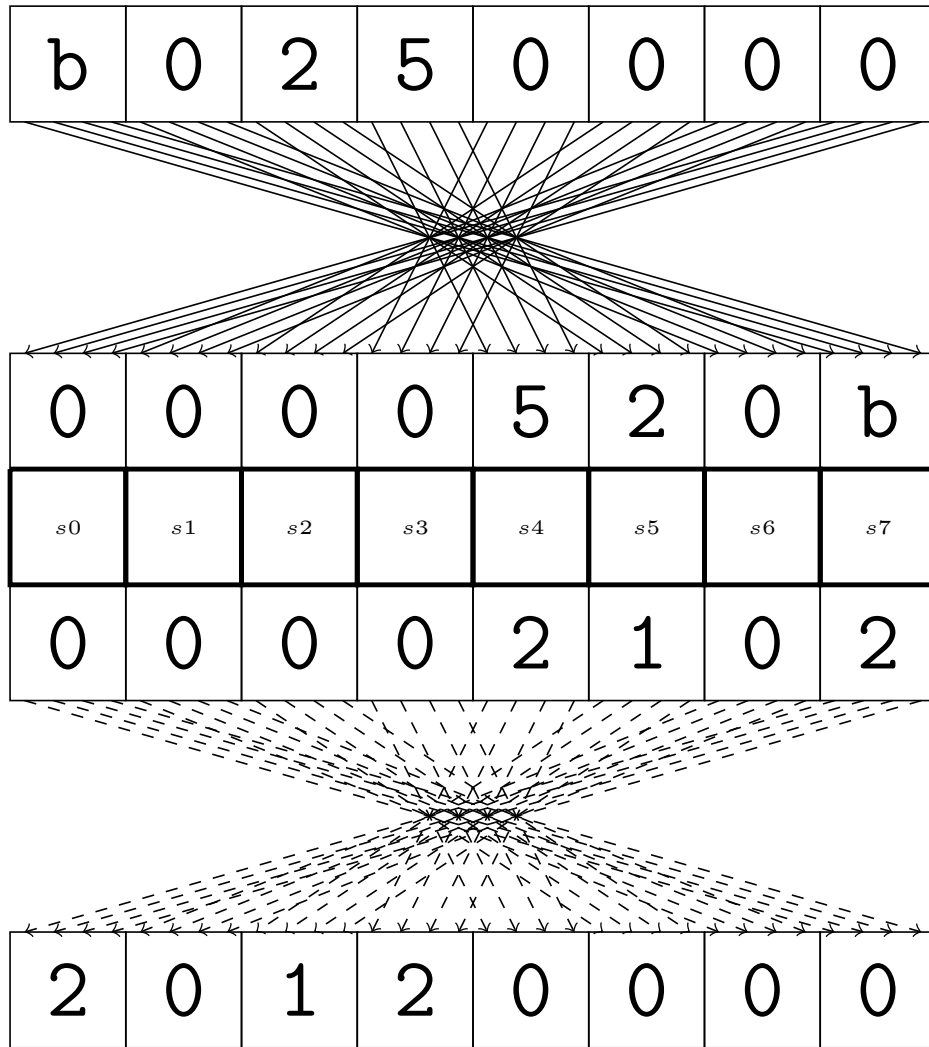
20 LBlock_round

Maximal differential probability: $2^{-6.0}$



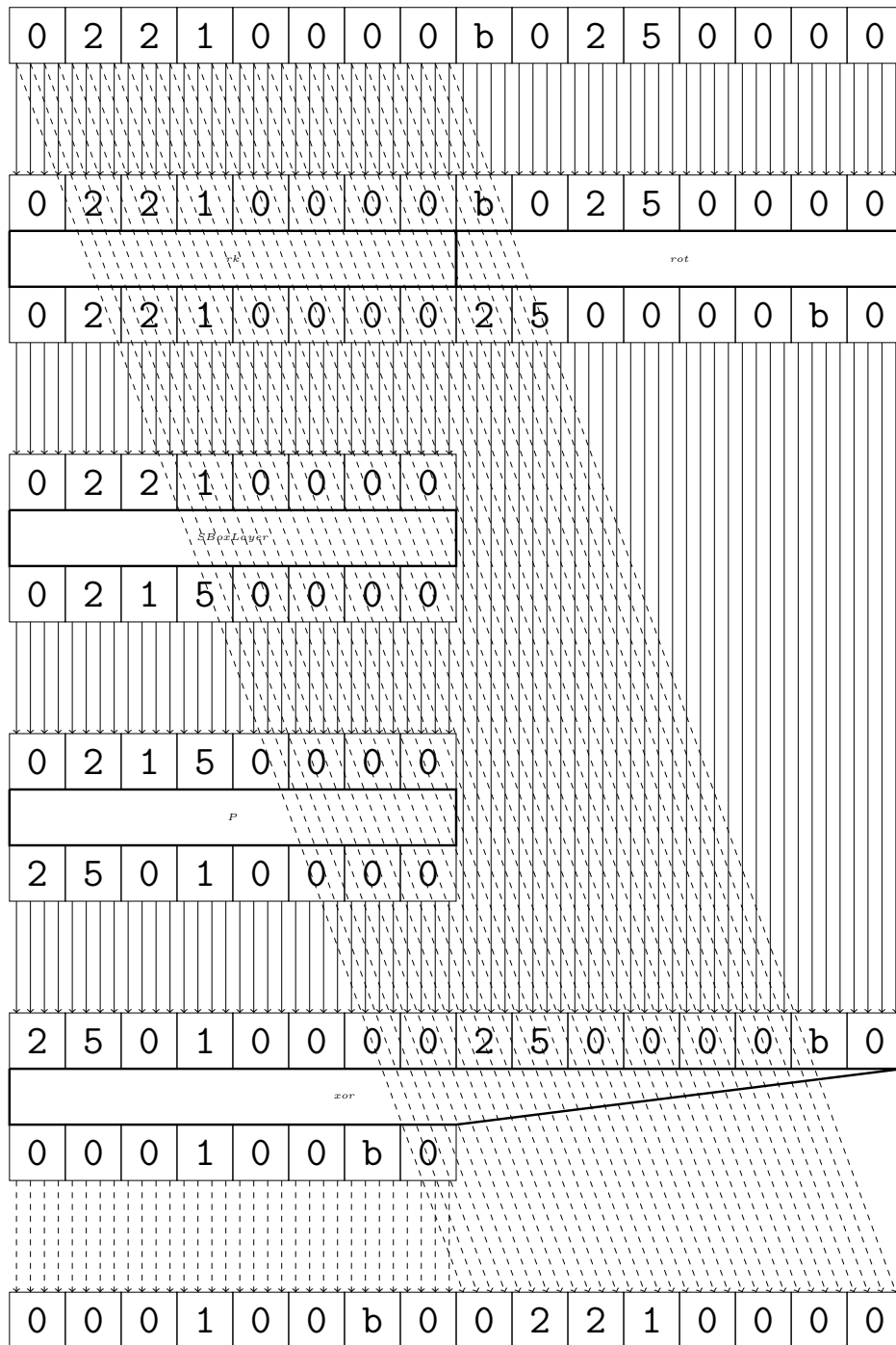
21 SBoxLayer

Maximal differential probability: $2^{-6.0}$



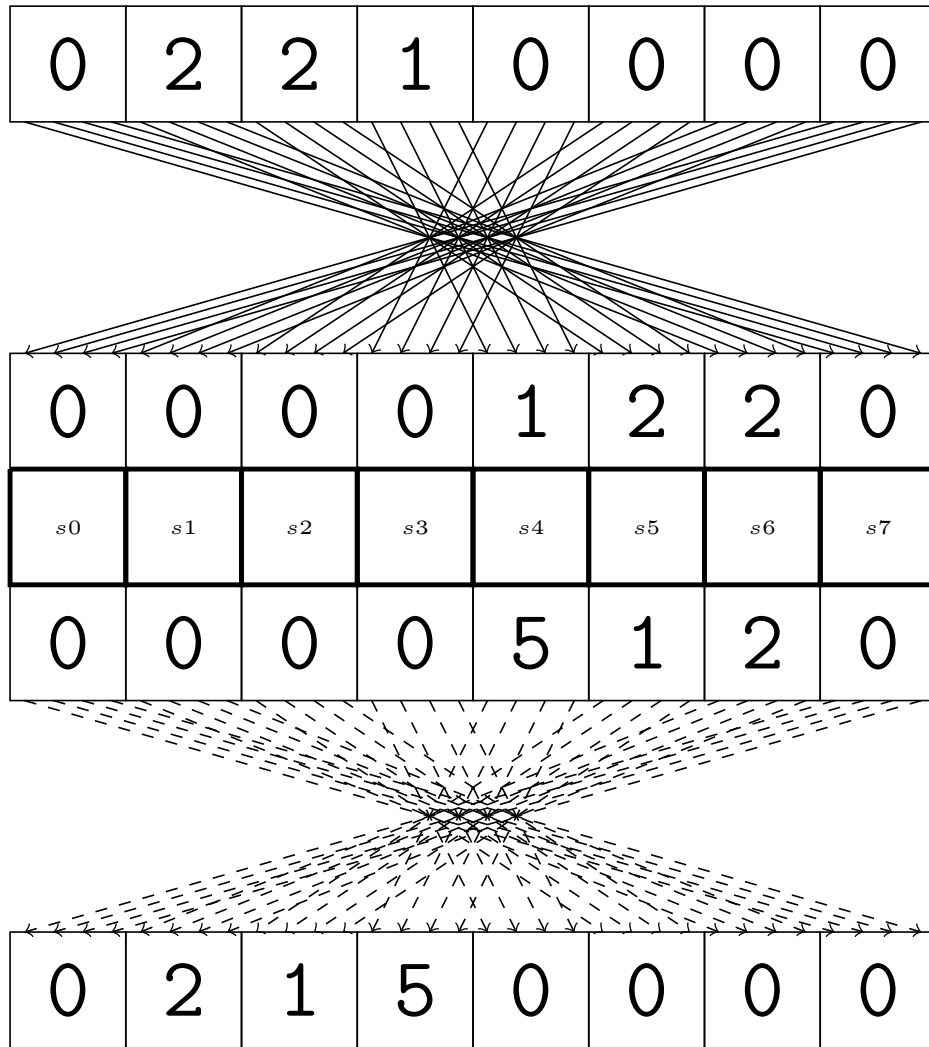
22 LBlock_round

Maximal differential probability: $2^{-6.0}$



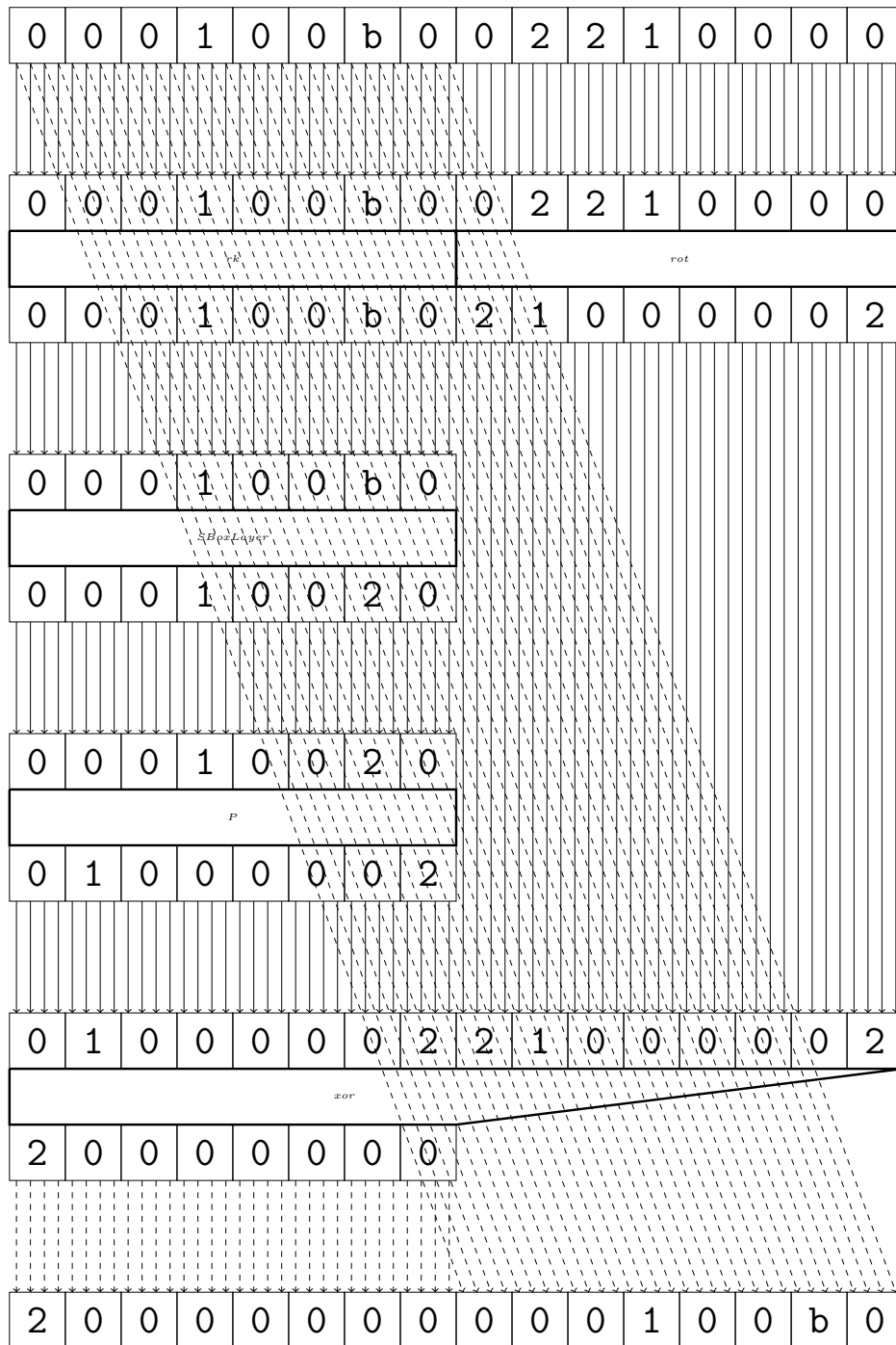
23 SBoxLayer

Maximal differential probability: $2^{-6.0}$



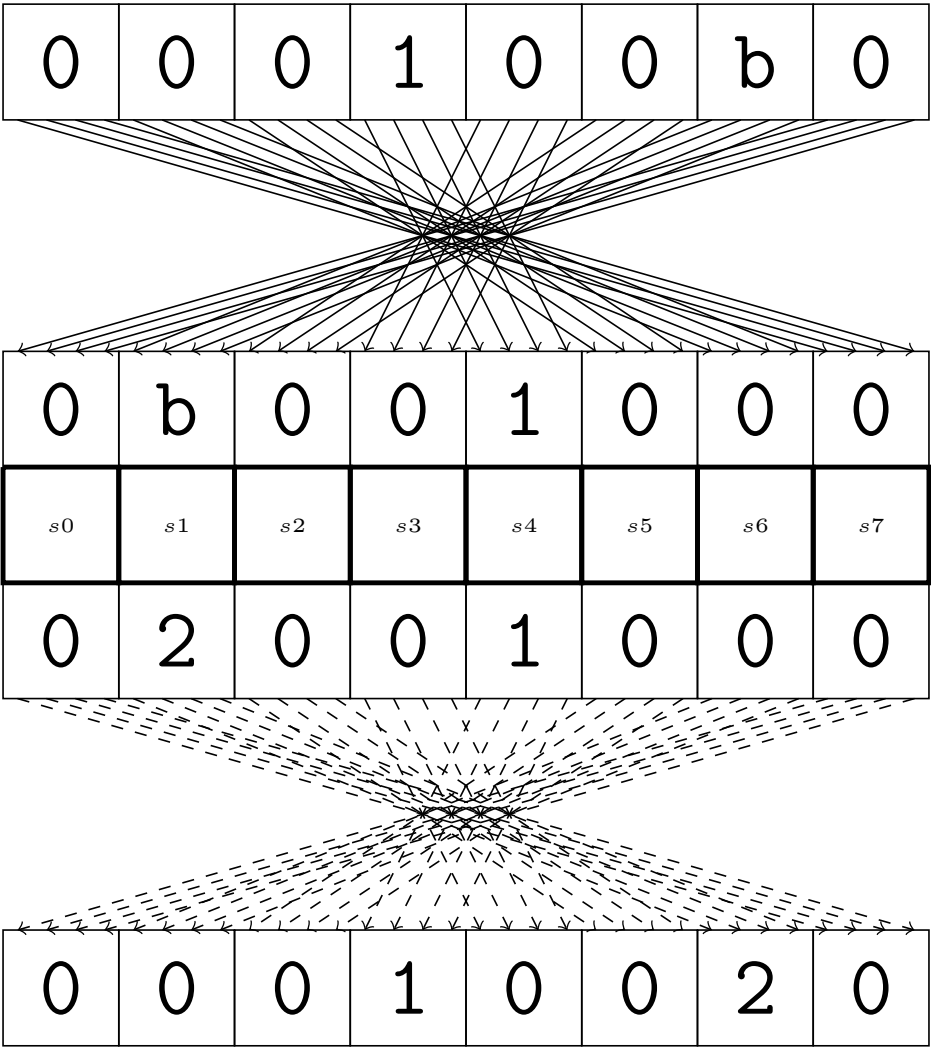
24 LBlock_round

Maximal differential probability: $2^{-4.0}$



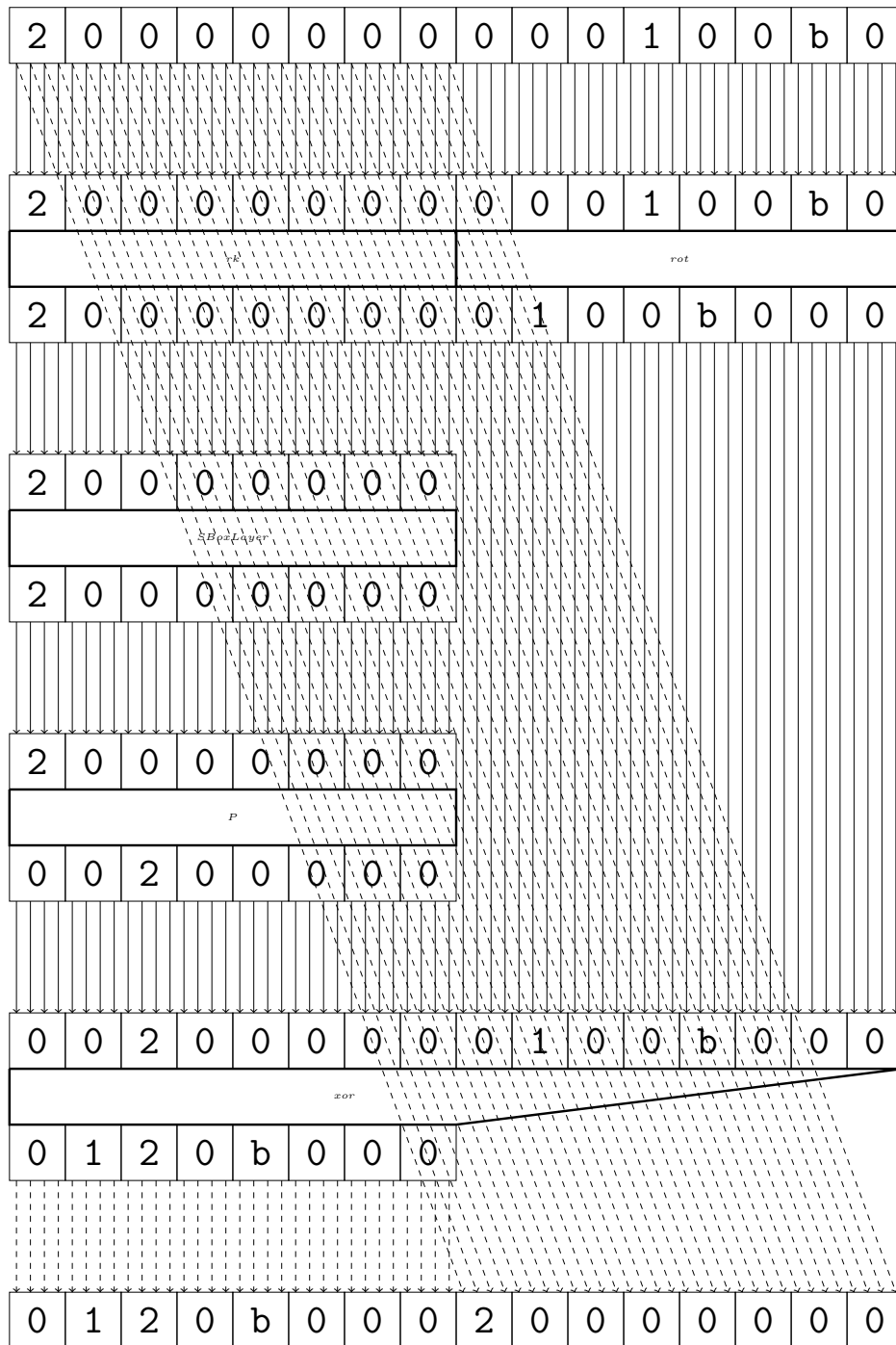
25 SBoxLayer

Maximal differential probability: $2^{-4.0}$



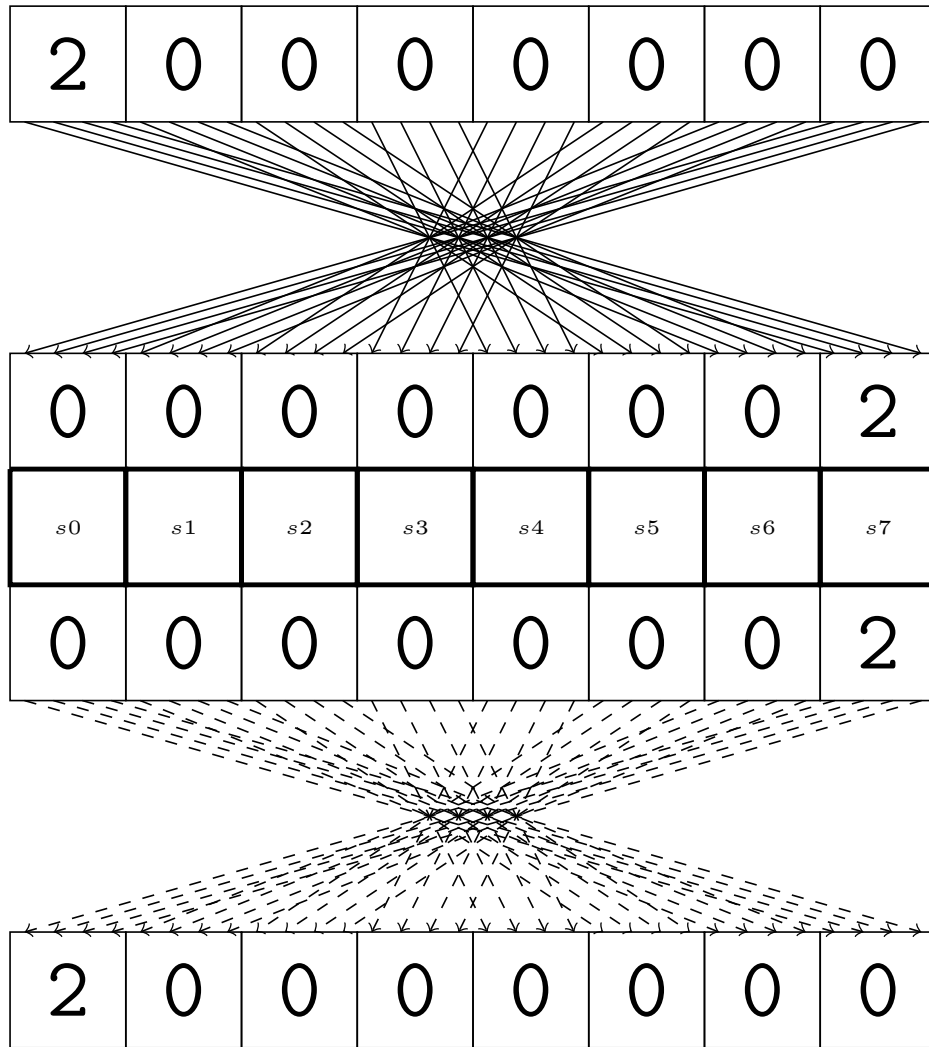
26 LBlock_round

Maximal differential probability: $2^{-2.0}$



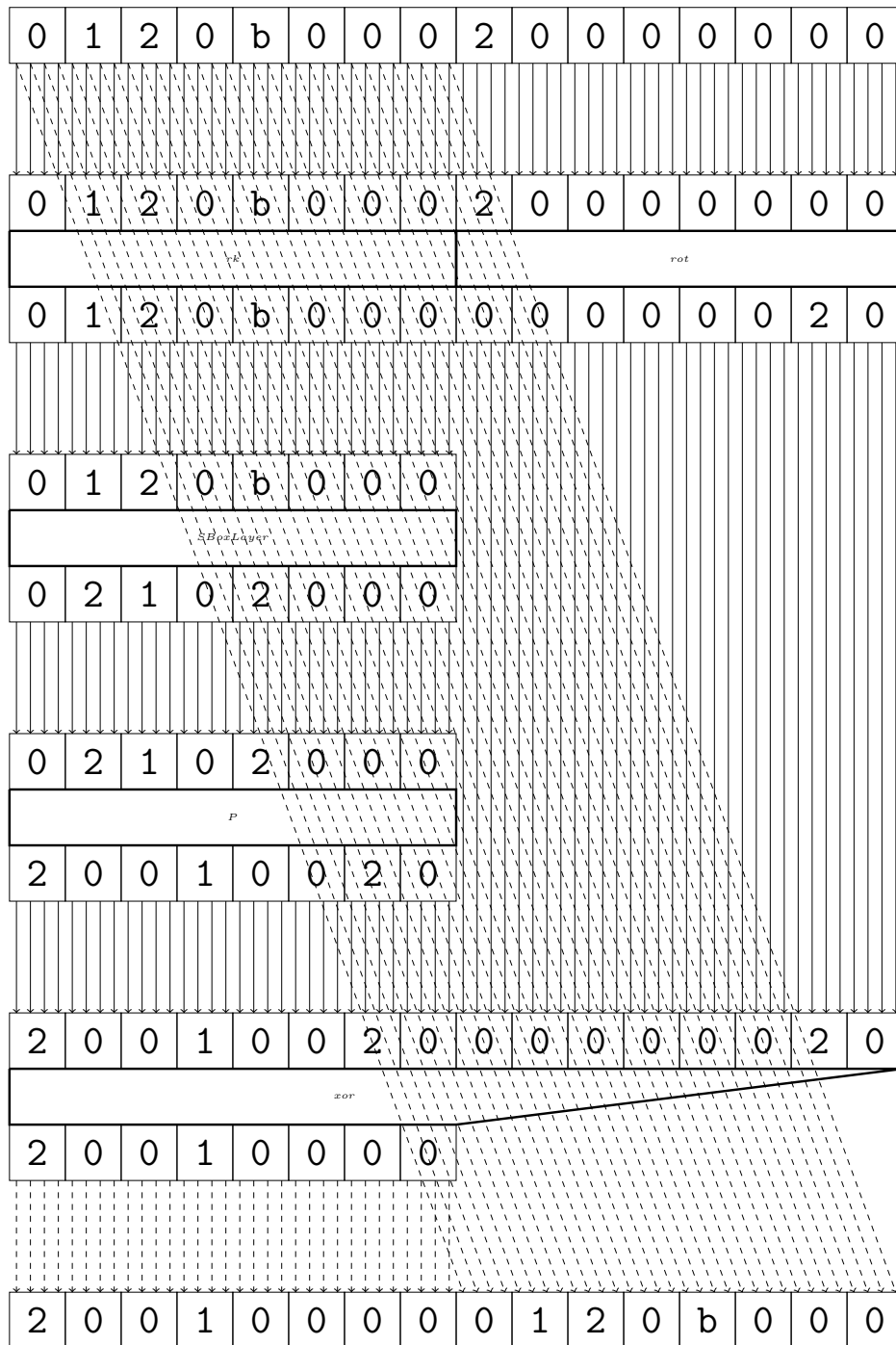
27 SBoxLayer

Maximal differential probability: $2^{-2.0}$



28 LBlock_round

Maximal differential probability: $2^{-6.0}$



29 SBoxLayer

Maximal differential probability: $2^{-6.0}$

